

An agent-built pipeline for in silico antisense oligonucleotide design, and what six adversarial reviewers found in it

A methodological report on ABCA4 c.161-395G>A

Sergio Mauricio Nuñez* Amyra Sanchez†

Implementation, pipeline execution and drafting were carried out by an AI agent (Claude, Anthropic)

2 August 2026

Abstract

The deep-intronic variant **ABCA4 c.161-395G>A** activates an aberrant pseudoexon (PE1b, 91 bp) and is the most prevalent deep-intronic allele in a reported Chinese Stargardt disease cohort. No antisense oligonucleotide (ASO) has been published or patented against it. We report a reproducible seven-module computational pipeline for in silico design of splice-switching ASO candidates of **PMO** chemistry, built end to end by an AI agent over six days, and then submitted to a panel of six independent adversarial reviewers.

The pipeline reproduces two results with external anchors. First, the variant installs the -2 consensus adenine of the donor motif over an intact GT dinucleotide (**GGG|GTAGGT** $\rightarrow$ **GAG|GTAGGT**), a model-free and threshold-free observation. Second, two independently trained neural predictors place their maximum at exactly $+1$ and -89 relative to the variant, yielding a predicted pseudoexon of **91 bp** that matches the minigene measurement of Peng et al. (2025) exactly.

We report the adversarial review as a primary result rather than as a limitations paragraph. The panel returned *reject with invitation to major resubmission* and identified seven critical issues. Correcting them changed the conclusions: a strand-orientation bug in the melting-temperature calculation had propagated four modules upstream and generated a finding, a design record and a published conclusion; a negative control with deliberately wrong tissue reproduced the “three-predictor agreement” exactly, showing that agreement to be uninformative; and a claim repeated across six documents for four days had never been verified against the literature.

No candidate has been synthesised or assayed. We argue that for agent-built research the adversarial review is not optional quality control but part of the method, and we quantify why.

Contents

1	Motivation	2
2	Provenance and division of judgement	2
3	Pipeline	2
3.1	Representing an ASO to a splice predictor	3
3.2	Why a Pareto front and not a weighted score	3

*Project conception, scope decisions, adversarial review commissioning, supervision. *YAIS Lab*. Contact: mauricio.nunez@yaislab.org

†Domain review, literature curation, manuscript review. *YAIS Lab*.

4 Results with external anchors	3
4.1 The variant installs the donor consensus adenine	3
4.2 The predicted pseudoexon size matches an independent experiment	3
5 The adversarial review as a result	4
5.1 A Module 4 bug that travelled four modules upstream	4
5.2 A wrong-tissue control that dissolved an argument	4
5.3 A claim repeated for four days without verification	4
5.4 A correction that recovered what the filter had hidden	4
6 What this pipeline does not do	5
7 Discussion	5
8 Availability	5

1 Motivation

Stargardt disease type 1 is the most common inherited macular dystrophy, caused by biallelic variants in **ABCA4**. A relevant fraction of pathogenic alleles are not coding variants but **deep-intronic variants** that activate cryptic splice sites, leading the spliceosome to treat an intronic stretch as an exon — a *pseudoexon* — and to introduce a premature termination codon.

Peng et al. (2025, *IOVS* 66(1):65) showed by minigene assay that **c.161-395G>A** activates three pseudoexons (PE1b 91 bp, PE1c 255 bp, PE1d 265 bp), with 76.4 % of the resulting mRNA incorrectly spliced. They assessed splicing effects with the SpliceAI web lookup and reported only that all variants exceeded a delta score of 0.2; they did not report cryptic site positions, pseudoexon coordinates, or a mechanistic account of the base change.

Splice-switching ASOs bind the pre-mRNA by base complementarity and **sterically block** access to a splice site without degrading the transcript. No published or patented ASO targets this variant — which motivates the work, but also means **there is no positive control** with which to validate a design pipeline end to end.

2 Provenance and division of judgement

We state this before the methods because it conditions how the rest should be read.

The implementation, pipeline execution and drafting were produced by an AI agent. The following were author decisions, not the agent’s:

- **The chemistry.** PMO was chosen as a scope decision, against the retinal literature precedent (2'-MOE/PS), and documented with its costs.
- **Commissioning the adversarial review**, on the explicit reasoning that an agent cannot audit work it produced.
- **Challenging the agent’s assertions.** Three of the highest-impact errors surfaced because the authors asked, not because the agent detected them — as did the fact that PMO force-field parameters do exist, after the agent had asserted they did not.

We consider this disclosure a methodological requirement rather than an acknowledgement.

3 Pipeline

Seven modules, each with its own reproducible runner and declared limitations.

#	Module	Question	Result
1	Sequence	Where is the variant?	Coordinate confirmed by two routes
2	Oligo-walk	Which windows?	381 candidates
3	Heuristic filter	Viable oligos?	381 $\rightarrow$ 276
4	Thermodynamics	Bind well, reachable?	276 $\rightarrow$ 78
5	Off-target	Resemble other genes?	78 annotated by severity
6	Neural splicing	Is the false site real?	Donor +1, acceptor -89
6b	ASO masking	Does each patch switch it off?	12 abolish the pseudoexon
7	Ranking	Which to synthesise first?	Pareto front: 3

3.1 Representing an ASO to a splice predictor

Splice predictors take a single sequence strand and have no representation of a bound ligand. A PMO does not cleave or modify the RNA — it covers it. We therefore replace the ASO window with N, which the one-hot encoding of both predictor families maps to the null vector: “no readable sequence information here”. This is a direct proxy for steric block.

Two alternatives were rejected: mutating the window introduces spurious motifs, and excising it shifts all downstream coordinates. Notably, `predict_scores` already pads with "N" $\times$ 5000 on each side, so blocks of N are the operational convention of these models rather than an invented input.

3.2 Why a Pareto front and not a weighted score

The three ranking dimensions — block strength, off-target safety, thermodynamic quality — have incomparable units. Averaging them requires positing an exchange rate (how many base pairs of off-target homology are worth 0.01 of block strength) that **no data in this project supports**, precisely because there is no positive control against which to calibrate it. A Pareto front requires no such rate. It returns a set rather than a winner, which is the honest output.

4 Results with external anchors

These two do not depend on any metric we defined.

4.1 The variant installs the donor consensus adenine

Counting matches against the donor consensus MAG|GTRAGT, the wild-type sequence 1 nt downstream of the variant reads GGG|GTAGGT and the mutant reads GAG|GTAGGT. The variant itself supplies the adenine at position -2 — one of the most conserved positions of the donor consensus — over an intact GT dinucleotide, with purines at +3 and +5. This requires no model, no threshold and no predictor.

4.2 The predicted pseudoexon size matches an independent experiment

SpliceAI local inference over a 10 kb window predicts a cryptic donor gained at +1 (0.194 $\rightarrow$ 0.560) and a cryptic acceptor at -89 (0.061 $\rightarrow$ 0.271). The implied pseudoexon is **91 bp**, identical to the PE1b measured by minigene in Peng et al. (2025) — computational prediction and in vitro measurement converging by fully independent routes.

Pangolin, a different architecture trained by a different group, places its maximum at *exactly* the same two positions, with neighbouring positions at $\sim 1e-5$. A deliberate attempt to break this result during adversarial review failed.

What does not follow. SpliceAI does not explain PE1c or PE1d: their implied acceptors score 0.016 and 0.042, far below any usable threshold. The model accounts for one of three reported pseudoexons, and we state this rather than omit it.

5 The adversarial review as a result

Six independent reviewers — splicing biology, statistics, reproducibility, ASO therapeutics, code integrity, and a hostile red team — were given a dossier of every claim with its evidence level and regeneration command, plus explicit rules of engagement requiring that each objection cite a file, line or number.

The verdict was *reject with invitation to major resubmission*. Seven critical issues were raised. Four were independently verified before acceptance. Their consequences:

5.1 A Module 4 bug that travelled four modules upstream

`Bio.SeqUtils.MeltingTemp.Tm_NN` with the `R_DNA_NN1` table documents that the argument must be the **RNA** strand; the code passed the ASO. The table is asymmetric, so the order changes the number.

Correcting it moved the Module 4 funnel from **44 to 16** candidates, with only 6 in common. More consequentially, the seven candidates that abolished the acceptor *without covering it* — the observation that had motivated the project’s verdict criterion, its own design record, and a published conclusion about predictor agreement — **did not survive**. A defect in the cheapest module had generated a finding four modules downstream.

5.2 A wrong-tissue control that dissolved an argument

The project had reported that three independently trained predictors, one of them retina-specific, selected the same exact candidate set, and presented this as cross-validation. The panel proposed the missing control: run a model trained on *deliberately* the wrong tissue.

It reproduced the same set exactly. The agreement reflects shared architecture and shared training annotations, not tissue-robust biology. The control cost six minutes of CPU and had been available since the weights were installed; nobody in the project had run it, because nobody was looking for a way to refute their own argument.

5.3 A claim repeated for four days without verification

The assertion that no nearest-neighbour thermodynamic parameters exist for PMO propagated through six documents from a single unverified inference. Checking it found: the claim is essentially correct for PMO, but the NN model *is* documented as applicable to uncharged oligomers; parameters *do* exist for 2'-MOE/PS, the chemistry of the actual retinal literature; and PMO force-field parameters *do* exist (Sarepta, 2023). Further, the experimental evidence shows phosphorodiamidate linkages *stabilise* the duplex, so the proxy **underestimates** the true melting temperature — a bias in the same direction as the inapplicable salt correction, so the two accumulate rather than cancel.

5.4 A correction that recovered what the filter had hidden

Because the melting-temperature threshold could not be interpreted for this chemistry, it was converted from a filter into an annotation — following a precedent the project had already set for off-target severity. The funnel went from 16 to **78** candidates.

The acceptor-only candidates **reappeared**: nine of them, with proxy melting temperatures of 36.5–40.1 °C, all below the discarded threshold. The mechanism was real; the buggy calculation had admitted them for the wrong reason and the corrected gate had hidden them entirely.

The final Pareto front returns to the same three candidates as the original analysis, now by a defensible route.

6 What this pipeline does not do

Stated with every result and not omissible when communicating it.

- **No experimental validation.** No ASO synthesised or assayed.
- **The masking proxy does not measure efficacy.** It is binary and total: 100 % occupancy, perfect block. Necessary, not sufficient.
- **PMO thermodynamics are a proxy of another chemistry**, now correctly oriented but still not PMO.
- **Off-target severity thresholds are uncalibrated** design choices.
- **Predictor agreement is not independent** and, per the tissue control, largely uninformative.
- **The pipeline is blind to splice-site displacement:** it scores four fixed offsets and discards the rest of the profile.

7 Discussion

The scientific contribution of this work is modest and bounded: a mechanistic account of one pseudoexon, computationally confirmed, plus a reproducible pipeline that produces a shortlist no one has tested.

The methodological contribution is, we think, larger. An AI agent produced a system with real engineering discipline — 224 tests, three documented environments, provenance files, archived pre-correction results — and simultaneously produced conclusions that did not survive contact with adversarial review. Crucially, **the failures were not arithmetic**. Six reviewers found no calculation error. They were failures of framing: which metric to define, which control not to run, which claim to repeat without checking.

That asymmetry is the finding we would emphasise. Agent-built research is likely to be strong where the work is mechanical and weak where it requires deciding what would refute you. Adversarial review is therefore not quality control appended at the end; for this mode of work it is part of the method.

8 Availability

Code, results and regeneration commands: <https://github.com/smnunez404/bio-oligonucleotidos> (MIT licence; generated data CC BY 4.0). Every number cited here has a regeneration command in the repository README. The adversarial review report and the claims dossier are maintained in the project’s documentation vault and are available on request.

Acknowledgements

We thank the authors of SpliceAI (Illumina), Pangolin (Zeng & Li), Retina-SpliceAI (Riepe et al., CMBI/Radboud UMC) and ViennaRNA for making their work openly available, and Peng et al. and Kaltak et al. for the experimental measurements that made external anchoring possible.